CereBro — Android Client

Engineering work report — audit, defect remediation and design-system consolidation

Repository pawan084/cerebrozen **Branch** redesign/light-dawn **Engineer** Abhimanyu Kumar **Date** 12 August 2026

70 Screens audited	**13** Defects found & fixed	**1** Crash eliminated	**16** Source files changed	**2,654** Lines added / removed	**464** Tests passing

1 · Repository synchronisation

Brought the feature branch up to date with upstream `main` across two separate synchronisation passes, resolving the branch divergence that had accumulated.

- **21 upstream commits** integrated into `redesign/light-dawn` — web app route work, landing trust pages, backend deployment documentation, Play Store listing assets, the Dawn palette consolidation, the unified mood taxonomy and the new `backend/app/services/moods.py` service.
- First pass resolved as a clean fast-forward; second pass required a merge commit once local work existed. No conflicts in either.
- Verified post-merge that no upstream commit remained unmerged, and confirmed the branch was pushed and in sync with the remote.

2 · Full codebase audit

Conducted a systematic audit of the entire Android client, cross-referencing the UI layer against the API client, the navigation graph, the localisation resources and the design-token system.

ANALYSIS PERFORMED	COVERAGE
Screen-by-screen data-source mapping	84 composables across 70 screens classified as API-backed, local-store-backed, or hardcoded
Navigation graph reachability	Every `composable()` route cross-referenced against every navigation call site
API endpoint coverage	All `Api.*` helpers traced to their call sites to find endpoints with no reachable UI
Localisation coverage	String literals vs `stringResource` counted per file; English and Hindi resource files compared
Design-token compliance	Raw hexadecimal colour usage located per file against the token system
Dead-code and dead-control detection	Empty click handlers, non-clickable controls and disabled code blocks identified
Git archaeology	Traced the regression to the specific commits that introduced it

Root cause established. A prior redesign pass had repointed nine navigation routes onto newly written design screens. A follow-up commit wired some of them to the API; the remainder shipped as static mockups, and the original working screens were left in the codebase, imported but unreachable.

Findings documented

- `ANDROID_AUDIT.txt` — a 435-line report covering every finding with file paths, line numbers, reproduction steps and proposed fixes, organised into eight sections including a “confirmed not a problem” section to prevent correct behaviour being “fixed” later.
- **14 API endpoints** identified as having no reachable user interface — habit tracking, per-item AI memory (view/edit/delete), personalised recommendations and goal completion — despite being fully implemented and tested server-side.
- **9 working screens** found orphaned; **8 navigation routes** found unreachable.
- Project documentation corrected where it made claims the code no longer supported.

3 · Defects found and resolved

ID	SEVERITY	DEFECT	RESOLUTION
`D-13`	Critical	Home screen rendered five hardcoded rows of fabricated activity — including a completed check-in that never happened — for every user on every launch, while the real plan, presence indicator, milestone line and recent activity were all disabled behind a permanently-false condition. The summary line above the list read real data, so the screen contradicted itself.	Both disabled blocks removed; fabricated rows deleted; genuine plan and presence content restored
`D-01`	Crash	Opening a check-in reminder from the notification inbox called navigation with a route name the graph never defined, throwing an uncaught exception	Route corrected; nudge destinations now validated against the same allow-list the deeplink resolver uses; regression test added
`D-10`	High	The thought-record screen's save control was a styled container with no click handler attached — written reflections were silently discarded, and were also lost on rotation	Route repointed to the implementation that composes and persists the entry
`D-11`	High	Body-scan practice displayed a static progress ring and a frozen counter above a play control with an empty handler — the practice never ran	Route repointed to the functioning narrated practice
`D-04`	High	Reminders screen held all switches in local state with hardcoded values and a save button whose body was empty; the notification inbox, which reads the preferences that screen was supposed to write, therefore always reported nothing scheduled	Route repointed to the implementation that schedules, requests permission and persists
`D-05`	High	Personal baseline wrote to a preferences key nothing in the application reads, so the insights screen's baseline comparison could never populate	Route repointed to the implementation the insights screen actually reads from
`D-07`	High	Goals screen omitted habit tracking entirely and offered no way to complete or release a goal; resolved goals disappeared with no way to retrieve them	Route repointed to the full implementation, restoring habits and goal lifecycle

ID	SEVERITY	DEFECT	RESOLUTION
D-12	Medium	The distress-tolerance skill was a static list of four cards rather than the step-by-step walkthrough, losing the per-step rationale	Route repointed to the walkthrough implementation
D-02	Medium	Daily plan screen displayed a perpetual loading state when no plan existed, because the failure branch could never execute	Consolidated onto the implementation with a real empty state and retry
D-03	Medium	The plan's primary action marked the next step complete instead of opening it	Resolved by the same consolidation
D-08	Medium	Three breathing-session settings were local state the session never read, so a disabled chime still sounded; one had no backing setting at all	Two switches wired to the persisted settings the session consults; the unbacked one removed rather than faked
D-06	Medium	Sleep insights passed a day count into a parameter the API treats as a row limit, so the selected window and the displayed statistics described different periods	Removed with the mockup that contained it
D-09	Medium	Users who had not signed in were shown network-failure messages on every server-backed screen	Documented with a proposed remedy; carried forward

Fabricated content removed

Beyond the defects above, several screens presented invented information as though it were the user's own data. These were removed rather than left in place, and deliberately **not** reimplemented — an absent feature is honest, a convincing fake is not:

- A pattern-detail screen presenting an invented behavioural observation with fabricated supporting statistics and a confidence rating.
- A goal-detail screen with a hardcoded progress chart, a title supplied by the navigation graph itself, and non-functional resume and delete controls.
- A post-practice reflection screen whose four options were not clickable, one of which promised a follow-up action that never occurred.
- A content-detail screen listing a duration and a saved state it had invented.
- **14 composables deleted** in total.

4 · Design-system consolidation

Twelve top bars unified into one component

The application had accumulated twelve independently written navigation bars with divergent heights, background opacities, padding and colour handling. Later copies had also lost the accessibility affordances the earliest ones carried.

Before	12 hand-rolled headers — heights of 66 and 76 units, three different opacities, two padding scales, back arrows tinted from raw hexadecimal values
After	A single shared component with 14 call sites and no hand-rolled headers remaining
Accessibility restored	Every control regained its button role, screen-reader description and press feedback
Safety	The crisis-support control now occupies identical screen position on every screen that carries it, satisfying the project's two-tap accessibility requirement

Design-token compliance

Raw hexadecimal colour values in screen files — a review-blocking defect under project standards, and the reason several screens rendered incorrectly under the dark theme — reduced from **209** → **19**.

FILE	BEFORE	AFTER
Practice library and family	117	0
Home screen	29	3
Explore screen	11	5
Sleep screen	10	4
Shared components, onboarding, misc.	42	7

5 · Localisation

The application ships a Hindi locale, but a substantial portion of the redesigned surfaces carried their copy as inline literals, which cannot be translated. The highest-impact case was addressed first.

Crisis support screen

All sixteen Hindi strings for this screen already existed in the resource file and were going entirely unused, because the route had been moved to a screen written with inline English. The one screen a person opens at their most vulnerable moment was English-only after they had explicitly selected Hindi — including the sentence stating that the product is not an emergency service. In addition, a regional label hardcoded the word “ENGLISH” beside helpline numbers whose own service answers in twenty languages.

- **60+ strings externalised** across the crisis, practice-library, breathing-preparation and gratitude screens, with Hindi translations authored for all of them.
- The practice family had **no Hindi resources at all** prior to this work — not a single key existed.
- Shared accessibility labels and top-bar subtitles added to both locales.

- A user-facing count that read “five” above six items corrected.

6 · Verification

CHECK	RESULT	NOTES
Unit test suite	464 passing	Two tests corrected: one asserted the crashing route value, pinning the defect rather than catching it
Android lint (debug)	Clean	Full build
Claims verification script	Clean	204 user-facing files checked for unsupported product claims
Kotlin compilation	Clean	Verified after each stage of the refactor

A new regression test was added binding notification destinations to the same validated route set the deeplink resolver uses, so the two paths by which an external string becomes navigation can no longer diverge.

7 · Deliverables

COMMIT	DESCRIPTION	SCOPE
`a316884f`	Android audit — full findings report	435 lines
`e1a18c2a`	Merge upstream main into feature branch	21 upstream commits
`41994d85`	Route consolidation, mock removal, crash fix, top-bar unification	13 files · +291 / −1,086
`c07447a5`	Localisation, design tokens, remaining header migration	7 files · +372 / −480

All work committed and pushed to the upstream feature branch. Project documentation (`docs/TODO.md`) updated with a full record of the remediation and the items deliberately left open.

8 · Carried forward

Recorded honestly in both the audit report and the project task list rather than left undocumented:

- Approximately 107 inline English strings remain across the Home and Explore surfaces and several smaller screens; these two will still display in English under a Hindi locale.
- 19 raw colour values remain outside the token system.
- A push-capability endpoint remains uncalled, and the corresponding settings control does not yet exist.
- A guest-mode empty state is specified but not yet implemented.
- The build has been verified by the automated suite only; it has not yet been exercised on a physical device or emulator.